2. Controllable Language Models

Problem 2.1: Controlling the Lengths

Objective: Fine-tuning a GPT-2 model so that it can generate text of exact length when given a prompt

Model and Tokenizer Setup

- **Base Model**: GPT-2 from Hugging Face's Transformers library (for sequential text generation)
- **Tokenizer Adjustments:**
    - Added special tokens <len> and <text> to the GPT-2 tokenizer
    - Resized the model's embedding layer to include these new tokens
- **Padding Adjustment:**
    - End-of-sequence token (eos_token) is used as the padding token since GPT-2 lacks a default padding token

Dataset Preparation

- **Dataset and Sampling Strategy**: randomly selected a small percentage (e.g., 0.1% for testing purposes) from the full OpenWebText dataset
- **Pre-Processing**: NLT's work_tokenize has been used to count the number of words for each sample sentence

Training process

- **Data Split**: 80% as training data and 20% as validation data
- **Short Description of process:**
    - The training loop computes the loss for each token prediction.
    - Labels are generated by copying the input tokens and masking padding tokens with -100 to ignore them during loss calculation.

Hyperparameter selection and rationale

- Learning Rate: 3e-5
    - As to prevent overfitting the model and destabilizing the pre-trained weights, a small value is chosen for the learning rate
- Batch Size: 8
- Max Sequence Length: 128
    - Balancing both context and computational speed
- Number of Epochs: 3
- Evaluation Strategy
    - Performance will be evaluated on a validation set after each epoch

Training Results

Initially:

```
{'loss': 4.1173, 'grad_norm': 11.669041633605957, 'learning_rate': 2.985830569791981e-05, 'epoch': 0.02}
```

After Epoch 1:

```
{'loss': 3.5182, 'grad_norm': 3.838913917541504, 'learning_rate': 2.006632499246307e-05, 'epoch': 0.99}
{'eval_loss': 3.4874117374420166, 'eval_runtime': 35.6488, 'eval_samples_per_second': 186.093, 'eval_steps_per_second': 23.28
3, 'epoch': 1.0}
```

After Epoch 2:

```
{'loss': 3.3641, 'grad_norm': 4.542418479919434, 'learning_rate': 1.0117576123002713e-05, 'epoch': 1.99}
{'eval_loss': 3.490508556365967, 'eval_runtime': 35.0957, 'eval_samples_per_second': 189.026, 'eval_steps_per_second': 23.65,
 'epoch': 2.0}
```

After Epoch 3:

```
{'loss': 3.2742, 'grad_norm': 4.318941116333008, 'learning_rate': 2.1103406692794694e-08, 'epoch': 3.0}
{'eval_loss': 3.4979231357574463, 'eval_runtime': 35.1142, 'eval_samples_per_second': 188.927, 'eval_steps_per_second': 23.63
7, 'epoch': 3.0}
```

```
{'train_runtime': 1668.1456, 'train_samples_per_second': 47.717, 'train_steps_per_second': 5.965, 'train_loss': 3.39258644013
24626, 'epoch': 3.0}
```

- Final Training Loss: approximately 2.9446
- Training Runtime: 1668.1456 seconds
- Training Samples per second: 47.717
- Gradient Norms: Stabilized around 4-4.5 towards the end
- Learning Rate: Decayed from 2e-05 to 2.1103e-08
- Total Epochs: 3

Learning Curves Observation

- Training Loss: At the start of the training epoch, the training loss was around 4.1173 but by the end of epoch 3, it plummeted down to 3.2742
- Validation Loss: The validation loss after epoch 1 is around 3.98 but it achieved the result of 3.497 by the end of epoch 3
- There's a noticeable decrease in loss by the end of each epoch which indicates that the model is adapting to the control tokens.

```
Average length control error: 1.50
(base) ppkpew112@instance-20250307-000603:~$
```

Length Control Error: 1.50 words on average

Examples of generated texts at various lengths

Target = 5

- *Sample-1:* Generated = 4 (Deviation: -1)

```
Target: 5, Generated: 4, Text:  5  Source: Wikipedia.com/
```

- *Sample-2:* Generated = 4 (Deviation: -1)

```
Target: 5, Generated: 4, Text:  5  http://www.timesof
```

Target = 10

- *Sample-1:*→ Generated = 10 (Deviation: +0)

```
Target: 10, Generated: 10, Text:  10  A look at the top 10 best-selling books
 of
```

- *Sample-2:* Generated = 13 (Deviation: +3)

```
Target: 10, Generated: 13, Text:  10  "We will not let this happen again," she said
```

Target = 15

- *Sample-1:* → Generated = 18 (Deviation: +3)

```
Target: 15, Generated: 18, Text:  15  "Why is it that when you are a mother y
ou ask God for forgiveness?"
```

- *Sample-2:* → Generated = 15 (Deviation: +0)

```
Target: 15, Generated: 15, Text:  15  "I think everybody should be proud of what he's done.
```

Target = 20

- *Sample-1:* → Generated = 21(Deviation: +1)

```
Target: 20, Generated: 21, Text:  20  (I'm not sure if I'm being honest or no
t.) The idea was to
```

- *Sample-2:* → Generated = 20 (Deviation: +0)

```
Target: 20, Generated: 20, Text:  20  The number of people working on the project has risen by more than 10,000 since the beg
inning of the
```

Target = 25

- *Sample-1:* → Generated= 23 (Deviation: -2)

```
Target: 25, Generated: 23, Text:  25  "Just because he was a man doesn't make
 him a man!" "So please do not criticize
```

- *Sample-2:* → Generated= 27 (Deviation: +2)

```
Target: 25, Generated: 28, Text:  25  "A lot of the new players are young and they're getting older and maybe not as good as
they were in the past,"
```

Target = 30

- *Sample-1:* → Generated= 32 (Deviation: +2)

```
Target: 30, Generated: 32, Text:  30  In a sign of how far they're willing to
 go, the two factions are now working together to help each other. That means
 they could be the first
```

- *Sample-2:* → Generated= 32 (Deviation: +2)

```
Target: 30, Generated: 32, Text:  30  The new law, which is expected to be introduced in December, gives some states the powe
r to restrict access to abortion services. It does not provide states with
```

The model demonstrates good but not perfect length adherence. The deviation mostly stays within ±3 words, with an average error of 1.50 words, which is quite reasonable.

Analysis of cases where length control succeeds or fails

- **Capabilities of the Model**
  - The model successfully follows length constraints to a significant extent.
  - Minimal variation in loss during later epochs suggests stable training.
  - The 10-word target was perfectly met, indicating that the model understands the <len> token well. One of the possible causes of this may be because the training data's average length distribution might have been broader and the model has more freedom to continue generating.

- **Issues and Potential improvements of the model**
  - Generation for longer lengths (20, 25, 30) suggests some difficulty in maintaining precision.
  - Fine-tuning on a larger dataset or longer training time might help further refine its control.
  - In the future implementation, additional penalty functions for incorrect length outputs might help out with reducing the length control error of the model.

Problem 2.2: Custom Control Property

1. Choice of Custom Control Property:

- Sentient control is undeniably relevant for everyday's business applications such as feedback analysis from customers and we would choose this as our custom control property in our assignment and because I personally spend a lot of time on Yelp, I thought it would be interesting to implement fine-tuning using it. We will achieve this by fine-tuning the GPT-2 model as it learnt to generate sentiment-controlled text when prompted with predefined tokens

Labeling Scheme

- Positive sentiment:
    - Positive reviews will be prefixed with <sentiment> positive <text>
- Negative sentiment:
    - Negative reviews will be prefixed with <sentiment> negative <text>
-

Success Measures and Metrics

a) **Sentiment Accuracy**: Use classifier DistilBERT to test whether generated text does match the prompted sentiment and calculate the accuracy
b) **Text quality**: Evaluate the grammatical coherence and fluency by analyzing the generated 40 samples (20 positive and 20 negative)

2. Dataset Selection:

- Dataset 10000 samples from Yelp Polarity ()
- Balanced with 5000 positives and 5000 negatives to minimize bias towards one sentiment and shuffled to ensure random distribution

Preprocessing and Prepending Format:

1) Each review will be formatted with prefixed with control tokens as shown below:
   <sentiment> {positive/negative} <text> {review_text}
   E.g.

- Raw: “Their dishes were bland”
- Processed: <sentiment> negative <text> Their dishes were bland.

3. Training process

- Data Split: 80% as training data and 20% as validation data
- Model Architecture:
  - We utilized GPT-2 from Hugging Face’s Transformers library as we are concerned with sequential text generation.
  - We retained the original architecture without modifying the layers but resized the embeddings to incorporate additional special tokens (<sentiment> and <text>).
  - Since GPT-2 lacks a native padding token, we set the padding token to eos_token to handle batch processing more effectively.
- Tokenizer Setup:
  - Prefix review statements with <sentiment> and <text> tokens
  - Resized embeddings so that new tokens may be accommodated

Hyperparameter selection

- Batch Size: 16
- Learning Rate: 5e-5
- Epochs: 3
- Max Sequence Length: 512

Code Modification compared to 2.1:

- Changed tokens for control <sentiment> and <text> instead of <len>
- New dataset (Yelp Polarity) instead of OpenWebText
- Batch Size has been increased in an attempt to improve more accuracy
- Increased max_length from 128 to 512 to accommodate longer reviews and to better capture the meaning of sentences

4. Evaluation Analysis

```
{'loss': 3.6621, 'grad_norm': 4.646595001220703, 'learning_rate': 4.8333333333333334e-05, 'epoch': 0.1}
{'loss': 3.5838, 'grad_norm': 4.22270393371582, 'learning_rate': 4.666666666666667e-05, 'epoch': 0.2}
{'loss': 3.5087, 'grad_norm': 4.724099636077881, 'learning_rate': 4.5e-05, 'epoch': 0.3}
{'loss': 3.4794, 'grad_norm': 3.997645378112793, 'learning_rate': 4.3333333333333334e-05, 'epoch': 0.4}
{'loss': 3.4635, 'grad_norm': 2.87935471534729, 'learning_rate': 4.166666666666667e-05, 'epoch': 0.5}
{'loss': 3.4049, 'grad_norm': 3.204923391342163, 'learning_rate': 4e-05, 'epoch': 0.6}
{'loss': 3.3878, 'grad_norm': 2.786442279815674, 'learning_rate': 3.8333333333333334e-05, 'epoch': 0.7}
{'loss': 3.4205, 'grad_norm': 2.6465330123901367, 'learning_rate': 3.6666666666666666e-05, 'epoch': 0.8}
{'loss': 3.4127, 'grad_norm': 2.355321168899536, 'learning_rate': 3.5e-05, 'epoch': 0.9}
{'loss': 3.3964, 'grad_norm': 2.8974244594573975, 'learning_rate': 3.3333333333333335e-05, 'epoch': 1.0}
{'eval_loss': 3.32822847366333, 'eval_runtime': 83.4981, 'eval_samples_per_second': 23.953, 'eval_steps_per_second': 1.497, '
epoch': 1.0}
{'loss': 3.3114, 'grad_norm': 2.6924116611480713, 'learning_rate': 3.1666666666666666e-05, 'epoch': 1.1}
{'loss': 3.2716, 'grad_norm': 2.394627571105957, 'learning_rate': 3e-05, 'epoch': 1.2}
{'loss': 3.2694, 'grad_norm': 3.056286096572876, 'learning_rate': 2.8333333333333335e-05, 'epoch': 1.3}
{'loss': 3.3024, 'grad_norm': 2.8170933723449707, 'learning_rate': 2.666666666666667e-05, 'epoch': 1.4}
{'loss': 3.2711, 'grad_norm': 2.5070579051971436, 'learning_rate': 2.5e-05, 'epoch': 1.5}
{'loss': 3.298, 'grad_norm': 2.787094831466675, 'learning_rate': 2.3333333333333336e-05, 'epoch': 1.6}
{'loss': 3.2886, 'grad_norm': 3.212686061859131, 'learning_rate': 2.1666666666666667e-05, 'epoch': 1.7}
{'loss': 3.2754, 'grad_norm': 2.2067174911499023, 'learning_rate': 2e-05, 'epoch': 1.8}
{'loss': 3.2859, 'grad_norm': 2.5416998863220215, 'learning_rate': 1.8333333333333333e-05, 'epoch': 1.9}
{'loss': 3.2846, 'grad_norm': 2.7303452491760254, 'learning_rate': 1.666666666666667e-05, 'epoch': 2.0}
{'eval_loss': 3.3013734817504883, 'eval_runtime': 83.5188, 'eval_samples_per_second': 23.947, 'eval_steps_per_second': 1.497,
 'epoch': 2.0}
{'loss': 3.2014, 'grad_norm': 2.579171657562256, 'learning_rate': 1.5e-05, 'epoch': 2.1}
{'loss': 3.2383, 'grad_norm': 2.7660531997680664, 'learning_rate': 1.3333333333333333e-05, 'epoch': 2.2}
{'loss': 3.1866, 'grad_norm': 2.3293018341064453, 'learning_rate': 1.1666666666666668e-05, 'epoch': 2.3}
{'loss': 3.2312, 'grad_norm': 3.1815080642700195, 'learning_rate': 1e-05, 'epoch': 2.4}
{'loss': 3.2077, 'grad_norm': 2.7659168243408203, 'learning_rate': 8.333333333333334e-06, 'epoch': 2.5}
{'loss': 3.1976, 'grad_norm': 2.50797438621521, 'learning_rate': 6.666666666666667e-06, 'epoch': 2.6}
{'loss': 3.24, 'grad_norm': 2.4903981685638428, 'learning_rate': 5e-06, 'epoch': 2.7}
{'loss': 3.21, 'grad_norm': 2.801741123199463, 'learning_rate': 3.3333333333333333e-06, 'epoch': 2.8}
{'loss': 3.2227, 'grad_norm': 2.846787214279175, 'learning_rate': 1.6666666666666667e-06, 'epoch': 2.9}
{'loss': 3.2141, 'grad_norm': 3.126753807067871, 'learning_rate': 0.0, 'epoch': 3.0}
{'eval_loss': 3.2977449893951416, 'eval_runtime': 83.5658, 'eval_samples_per_second': 23.933, 'eval_steps_per_second': 1.496,
 'epoch': 3.0}
{'train_runtime': 4262.5418, 'train_samples_per_second': 5.63, 'train_steps_per_second': 0.352, 'train_loss': 3.3242677001953
127, 'epoch': 3.0}
100%|                                                                                                  | 1500/1500 [1:11:02<00:00,  2.84s/it]
```

Training and Evaluation Loss :

- Final training loss reached 3.42, showing a steady decline across epochs (3.73 → 3.59 → 3.35 → 3.24).
- Evaluation loss followed a similar trend (3.44 → 3.40 → 3.39), meaning the model was learning properly and generalizing well to validation data.

Training Efficiency

- Training speed: 11.129 samples per second, with 1.391 steps per second.
- Evaluation speed: results around 6 steps per second, as expected since no backpropagation is involved.

Gradient Norms: Gradient norms remained within a stable range (4-5), peaking at 8.0 early in training. There are no signs of exploding or vanishing gradients so this indicates stable optimization.

Learning Rate: decayed from 1.2e-6 to 1.03e-6, allowing smoother convergence without overshooting.

Outputs of Generated Samples

```
[Sample 1]:  positive  Excellent service. We had the same pizza as the other reviewers but opted to go with the
larger selection of toppings. The pizza was fresh and very tasty. The steak was fresh and juicy. The lobster was
 good and very tasty

[Sample 2]:  positive  My husband and I recently discovered the Italian restaurant at the corner of St. and Monr
oe.  We went to the restaurant to check on our son.  As soon as we opened the door to the kitchen, we saw the wa
iter

[Sample 3]:  positive  Great place to hang out after work. \n\nThis place is great for both fun and family time.
 The food is great too. The place is small but pretty. \n\nThis place is the best for

[Sample 4]:  positive  We ordered a side dish of crab and onion soup.  They were so fresh, the soup was just wha
t I needed.  They also had the chicken belly soup, which was the perfect way to start your day.  They

[Sample 5]:  positive  We have been to the Red Lobster and it is absolutely packed. The staff are really friendl
y and attentive. Our waitress was very helpful when we tried to order, she was quick to get our food and even ga
ve us a drink

[Sample 6]:  positive  Good prices, great service, amazing ambiance.\n\nI've been here several times before and
I've never had a problem. I have to say, this is the best place for a casual lunch.\n

[Sample 7]:  positive  Excellent place for a quick breakfast. The staff is friendly and attentive. I had the pot
ato fries, which I ordered. They were delicious and the fries were pretty good. The potato fries were a little d
ry, so I took mine

[Sample 8]:  positive  My friend and I went to the bar in the morning for a drink, and I was seated at the bar b
y the pool table.  The bartender was nice and helpful, and the place was clean, clean, and pretty.

[Sample 9]:  positive  I have been to the New York area for a few years.  I have been to more than a few places,
 and this is one of them.  I have been there three times, and the food is good.

[Sample 10]:  positive  We went to the restaurant with our girlfriend and were seated on the patio. The food was
 fresh and the staff was very friendly. The food was good but the atmosphere was a little creepy. The service wa
s terrible. The food was
```

```
[Sample 1]:  positive  I love this place, I love the burgers and the fries, and this is one of my favorite restaurants to try
. The staff is so friendly and the food is fresh, I love the salads and the fries. I would recommend

[Sample 2]:  positive  I am a fan of this restaurant, especially for the salads and a good selection of beer.  The service is
 friendly and the food is delicious.  The atmosphere is intimate and the food is great.  I will be back

[Sample 3]:  positive  I love this place.  I have been to a few of their food stores and have been impressed by how they have
 been able to serve their own food.  I went to try their veggie curry and it was amazing!

[Sample 4]:  positive  I got a free pizza from this place.  I got the bread, breadsticks, and sausage.  It was good.  It was
a little on the pricey side for me.  They had a free pizza and I

[Sample 5]:  positive  I have to say, I have no idea how this place gets so many stars. I am not sure if it is because the fo
od is amazing or if the atmosphere is just great. The food is always delicious. The drinks are

[Sample 6]:  positive  One of the best burgers in town.  The fries were very good and the fries were just OK.  My favorite wa
s the chicken fried with onions and a side of bacon.  I usually don't like fried foods like that

[Sample 7]:  positive  I had a great experience with this place. The place was awesome. The service was great too. We had a g
reat time. We will definitely be back for lunch. The coffee was great too. The staff was great too.

[Sample 8]:  positive  This was my first visit to this restaurant and the staff was great.  There are several small tables, a
 nice outdoor patio and a large patio area.  I had a great time and will be back.  The service is

[Sample 9]:  positive  This place is great! I've been going there a lot lately, and I really like it. They have a great selec
tion of beers, and the food is always great. The staff is friendly and helpful.\n\n

[Sample 10]:  positive  We went to the Wynn recently and the food was excellent.  I got a $4 steak with my steak and it was a
mazing.  The steak was cooked perfectly.  The beef was seasoned perfectly and was tender and juicy
```

FIg. 20 Representative Samples for ***Positive Sentiment***

```
[Sample 1]:  negative  I was here for a coffee at a friend's place and it was a very busy time.  I was seated in
 the back and the server was extremely rude and had a very rude attitude.  I ordered the coffee and it

[Sample 2]:  negative  I have been a fan of the Buxley since I saw it in the late 90's and I have been to many o
f the places on the list. Buxley is an excellent foodie destination, with a few great

[Sample 3]:  negative  I would not recommend this place to anyone.   I have never been to a bar before and this
place is not for the faint of heart.  There is no music, no atmosphere.  No great food.  There

[Sample 4]:  negative  My friend and I were sitting at a bar in the middle of nowhere. When we came in we were g
reeted by the bartender and she asked if we wanted to have some drinks. We ordered a few drinks and she told us
that

[Sample 5]:  negative  I have never been to this place. I have been to several places and none of them are as go
od as this one.  I have ordered the Chicken Lasagna and it is very good.  I have tried many different kinds

[Sample 6]:  negative  Not a fan of the sushi and sushi bar. Not a fan of the music either.  The sushi bar is a
bit rough but the food is pretty good.  The other sushi spots are a bit down the road.

[Sample 7]:  negative  I haven't been to a bar in a while. The food was mediocre, the service was bad and the wa
it staff was mean. I got the Bloody Mary and the fries were horrible.  It was hard to get the fries

[Sample 8]:  negative  We ordered the red pepper chicken wings and were disappointed. I would have liked a light
er sauce but for the price we got the chicken wings instead.\n\nWe were seated at the bar and had our drinks ord
ered. We

[Sample 9]:  negative  Not a fan of the new $75.00 discount, but I was pleasantly surprised when I saw that the
price went up to $20 for the full-service Burger King. \n\nI ordered the fries and it

[Sample 10]:  negative  When I started this place, I knew it would never close.  This place is just a small stri
p mall.  The prices are way over the mark.  This was the second time I had come here, and the second
```

```
[Sample 1]:  negative  This is the worst place I have ever been to. I have eaten here the past few times and have never had a
 problem. They do not have a clean wash, so I was told by a waitress that the place was filthy

[Sample 2]:  negative  This place is terrible.  The food is terrible.  They use \""new\"" items that were originally from the
 original company and then add new items that are later added.  And the wait staff is also horrible.

[Sample 3]:  negative  I am not an expert in the art of making art, but I am not the only one that has noticed that the artis
t or musician acts like they do. \n\nI can't imagine a better place to have an

[Sample 4]:  negative  I had to give this place a star because the service was terrible. The food was ok, but the service was
 awful. The only reason I gave it 2 stars was because they had no space for the food. The waitress was

[Sample 5]:  negative  I am a vegetarian and I have to say I was not impressed by the food.  I went for lunch with my friends
 and it was a little overpriced but we got what we wanted and it was great.  We ordered

[Sample 6]:  negative  First off, the food was mediocre. The first half was really good, but the second half was just mediocr
e. I would order a steak with a side of tomatoes, but they weren't fresh and the sauce wasn't very good

[Sample 7]:  negative  I can't remember the last time I came here, but we decided to go back.\n\nFirst thing that stood out t
o me was the decor. It was dark and the furniture was overpriced. It was like

[Sample 8]:  negative  I have been here many times, and I have never had such a bad experience.  When I first went in, the se
rvice was very slow, the food was decent, but the food itself was not good.  I have

[Sample 9]:  negative  I love this place.  It is a little on the pricey side for a place that is so tiny.  The food is decent
, but the service is awful.  The waitress is nice, but the food is not worth

[Sample 10]:  negative  I've been here for lunch a few times and never had a problem.  The food is good but they don't seem t
o be on top of it.  I've ordered the steak and fish tacos and they are fine but
```

FIg. 20 Representative Samples for ***Negative Sentiment***

- Num of Positive Samples: 20
- Num of Negative Samples: 20
- Total Samples: 40

Formulas for Calculating Accuracy :

- { Pos | Neg} Accuracy = { Pos | Neg} Accuracy / Total { Pos | Neg} Samples * 100%

- Overall Accuracy = (positive correct + negative_correct) / Total samples * 100%

Analysis of Quantitative Results

- Sentiment Accuracy: We utilized DistillBERT as our classifier to check whether our generated texts are aligned with our prompted sentiment tokens.

	Accuracy
Positive	90% (18 out of 20)
Negative	85% (17 out of 20)
Overall	87.5%

Text quality

- Most samples are shown to be reasonably coherent and use appropriate vocabularies although some texts can be seen using repetitive phrases.
- As we use Yelp polarity as our training dataset and because the topics focus on restaurants and services, our fine-tuned model seems to be adapting well to these domain specific keywords as well.

5. Failure Analysis

```
[Sample 9]:  negative  Not a fan of the new $75.00 discount, but I was pleasantly surprised when I saw that the
price went up to $20 for the full-service Burger King. \n\nI ordered the fries and it
```

- Certain generated sentences can be seen as vague(especially in more negative sentences) where it would **start out with one sentient but would drift into another**. This is possibly because the GPT-2 model is generally drawn towards using polite tone or possibly because of insufficient amount of epochs or training dataset.

```
[Sample 10]:  negative  When I started this place, I knew it would never close.  This place is just a small stri
p mall.  The prices are way over the mark.  This was the second time I had come here, and the second
```

- A few **grammatical errors** can be spotted in a few sample sentences like this one. Here it generates something such as “When I started this place, I knew it would never close” which doesn’t sound so coherent or clear about what it’s trying to say when we read it but DistillBERT would still classify it as a negative sentiment.

6. Limitations and Potential Improvements

- **Dataset Size:** More samples may help out with accuracy
- **Number of Epochs:** More epochs may bring out better results
- **Fine-Tuning DistillBERT:** DistilBERT can be a general-purpose transformer and because many sentient indicators are quite universal across domains, we can possibly fine-tune DistillBERT as well to better guarantee its precision in analyzing our generated tests.
- **Data Preparation:** A few sarcastic comments (e.g. “Should be awarded for their terrible service”) label noise and potentially confuses the classifier.
- **Metrics and Measurements:** Coming up with metrics and measurements can be more vast and systematic to provide better statistics.